Problem 1 Solution

The augmented matrix is:

$$\left[\begin{array}{ccc|c} 1 & -1 & 1 & 3 \\ -1 & -2 & c & 0 \\ 1 & c & -2 & 0 \end{array}\right]$$

Row reduce:

$$R_2 \leftarrow R_2 + R_1, \quad R_3 \leftarrow R_3 - R_1 :$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 1 & 3 \\ 0 & -3 & c+1 & 3 \\ 0 & c+1 & -3 & -3 \end{array}\right]$$

$$R_3 \leftarrow R_3 + \frac{c+1}{3} R_2 :$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 1 & 3 \\ 0 & -3 & c+1 & 3 \\ 0 & 0 & \frac{(c+4)(c-2)}{3} & c-2 \end{array}\right]$$

The last equation: $\frac{(c+4)(c-2)}{3} x_3 = c - 2$.

- If $c \neq -4$ and $c \neq 2$: unique solution. - If $c = 2$: infinite solutions. - If $c = -4$: no solution.

For $c = 2$, the reduced system:

$$\begin{cases} x_1 = 2, \\ x_2 - x_3 = -1. \end{cases}$$

Let $x_3 = t$, then $x_2 = t - 1$. Solution set: $\{(2, t-1, t) \mid t \in \mathbb{R}\}$.

Answer: (a) $c = -4$. (b) $c \neq -4$ and $c \neq 2$. (c) $c = 2$. Solution set: $\{(2, t-1, t) \mid t \in \mathbb{R}\}$.

Problem 2 Solution

(a) Apply row operations:

1. $R_2 \leftarrow R_2 + R_1$, $R_3 \leftarrow R_3 - 2R_1$:

$$\begin{bmatrix} 1 & 2 & 2 & 3 \\ 0 & 0 & 3 & 6 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

2. $R_2 \leftarrow \frac{1}{3} R_2$:

$$\begin{bmatrix} 1 & 2 & 2 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

3. $R_1 \leftarrow R_1 - 2R_2$:

$$\mathrm{RREF}(A) = \begin{bmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

(b)Record all the row operations used to transform A into R , and write down the corresponding elementary matrices. The matrix E is the product of these elementary matrices, with the order from left to right corresponding to their application on A from right to left.

$$E_1 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}, \quad E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad E_4 = \begin{bmatrix} 1 & -2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$E = E_4E_3E_2E_1 = \begin{bmatrix} 1 & -2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad R = EA$$

(c) From RREF:

$$\begin{bmatrix} 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = 0$$

Pivot columns: 1 and 3; free variables: $x_2 = s,\ x_4 = t$.

$$x_3 + 2x_4 = 0 \Rightarrow x_3 = -2t, \qquad x_1 + 2x_2 - x_4 = 0 \Rightarrow x_1 = -2s + t$$

$$\mathbf{x} = \begin{bmatrix} -2s+t \\ s \\ -2t \\ t \end{bmatrix} = s \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

$$\text{Solution set} = \text{Span} \left\{ \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \end{bmatrix} \right\}$$

(d) From RREF, pivot columns are 1 and 3, so take columns 1 and 3 of A:

$$\text{col1} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}, \quad \text{col3} = \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix}$$

Basis: $\left\{ \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix} \right\}$.

(e) Rank of A = number of pivot positions in RREF(A).

$$\text{rank}(A) = 2$$

Problem 3 Solution

Form the matrix with these vectors as columns

$$A = \begin{bmatrix} 1 & 0 & 0 & -1 \\ -1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & -1 & 1 & 0 \end{bmatrix}$$

Row operations:

$$R_2 \leftarrow R_2 + R_1 : \quad \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & -1 & 1 \\ 0 & -1 & 1 & 0 \end{bmatrix}$$

$$R_4 \leftarrow R_4 + R_2 : \quad \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

$$R_4 \leftarrow R_4 + R_3 : \quad \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

There are 3 pivots in the first 3 columns. The first 3 columns are linearly independent. The last column is linear combination of the first 3 columns. Hence, a basis for the minimal subspace is:

$$\left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ -1 \\ 1 \end{bmatrix} \right\}$$

Problem 4 Solution

(a) Since $(u + v) - 1(u + v) = 0$, the vectors $u + v, u + v$ are linearly dependent; thus they do not form a basis.

(b) Let us show that $u + v, v$ are linearly independent. Suppose

$$c_1(u + v) + c_2 v = 0.$$

Then

$$c_1 u + (c_1 + c_2)v = 0.$$

Since u and v are linearly independent, we obtain that $c_1 = 0$ and $c_1 + c_2 = 0$. This implies that $c_1 = c_2 = 0$, which shows that $u + v, v$ are linearly independent.

(c) Let us show that $u+v, u-v$ are linearly independent. Suppose

$$c_1(u+v)+c_2(u-v)=0.$$

Then

$$(c_1+c_2)u+(c_1-c_2)v=0.$$

Since u and v are linearly independent, we obtain

$$c_1+c_2=0$$

$$c_1-c_2=0.$$

Solving the last system of linear equations, we obtain $c_1=c_2=0$.

Remark: vectors u, v are linearly independent if and only if $u+cv, v$ are linearly independent for every c (the same argument as in 2). Similarly, u, v, w are linearly independent if and only if $u+cv, v, w$ are linearly independent for every c; and so on.

Problem 5 Solution

(a) We have: $(2a-1)v_1+(-1-b)v_2=0$. Since v_1, v_2 are linearly independent, we obtain $2a-1=0$ and $-1-b=0$. Answer: $a=\frac{1}{2}, b=-1$.

(b) The vectors v_1+3v_3, $2v_1+kv_3$, $2v_2$ are linearly dependent if and only if there exist scalars c_1, c_2, c_3, at least one of which is not zero, such that:

$$c_1(v_1+3v_3)+c_2(2v_1+kv_3)+c_3(2v_2)=0.$$

Expanding this expression, we get:

$$(c_1+2c_2)v_1+(2c_3)v_2+(3c_1+kc_2)v_3=0.$$

Since v_1, v_2, v_3 are linearly independent, the above linear combination equals zero if and only if all coefficients are zero:

$$\begin{cases} c_1+2c_2=0 \\ 2c_3=0 \\ 3c_1+kc_2=0 \end{cases}$$

We now seek all values of k for which this homogeneous system has a non-trivial (non-zero) solution. Applying the determinant test to the coefficient matrix:

$$\det\begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 2 \\ 3 & k & 0 \end{bmatrix}=0,$$

we compute the determinant and obtain:

$$k=6.$$

Problem 6 Solution

Answer: (a) no, (b) no, (c) no.

(a) $T\begin{bmatrix}0\\0\end{bmatrix} = \begin{bmatrix}0\\21\end{bmatrix} \neq \begin{bmatrix}0\\0\end{bmatrix}$, hence T is not a linear transformation. (If T was a linear transformation, then we would have $T\begin{bmatrix}0\\0\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix}$.)

(b) $K\begin{bmatrix}2\\2\end{bmatrix} = \begin{bmatrix}4\\10\end{bmatrix} \neq 2K\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}4\\8\end{bmatrix}$.

(c) $S\begin{bmatrix}1\\0\end{bmatrix} + S\begin{bmatrix}-1\\0\end{bmatrix} = \begin{bmatrix}0\\2\end{bmatrix} \neq \begin{bmatrix}0\\0\end{bmatrix}$. (If S was a liner transformation, then we would have $S\begin{bmatrix}1\\0\end{bmatrix} + S\begin{bmatrix}-1\\0\end{bmatrix} = S\begin{bmatrix}0\\0\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix}$.)

An example of a linear transformation:

$$N\begin{bmatrix}x\\y\end{bmatrix} = x\begin{bmatrix}1\\2\end{bmatrix} + 7\begin{bmatrix}x-y\\y\end{bmatrix}$$

because

$$N\begin{bmatrix}x\\y\end{bmatrix} = \begin{bmatrix}8 & -7\\2 & 7\end{bmatrix}\begin{bmatrix}x\\y\end{bmatrix}$$

(the standard form).

Problem 7 Solution

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. Let $\mathcal{B} = \{v_1, v_2, \cdots, v_k\} \subset \mathbb{R}^n$ and $T(\mathcal{B}) = \{T(v_1), \cdots, T(v_k)\}$.

(a). Assume that T is one-to-one. Show that if $\mathcal{B}$ is linearly independent, then $T(\mathcal{B})$ is linearly independent set of size k.

Solution: Since T is one-to-one, it maps distinct inputs to distinct outputs. Thus, $T(\mathcal{B})$ must contain k vectors.

To show that $T(\mathcal{B})$ is linearly independent, we need to show that the vector equation

$$x_1T(v_1) + x_2T(v_2) + \cdots + x_kT(v_k) = 0 \tag{0.1}$$

has only the trivial solution $x_1 = x_2 = \cdots = x_k = 0$. Since T is linear, we can write it as

$$T(x_1v_1 + x_2v_2 + \cdots + x_kv_k) = 0.$$

Since $T(0) = 0$ (see Lemma 2.1) and T is one-to-one, it follows that

$$x_1 v_1 + x_2 v_2 + \cdots + x_k v_k = 0.$$

If $\mathcal{B}$ is linearly independent, then this equation has only the trivial solution $x_1 = x_2 = \cdots = x_k = 0$. Consequently, (0.1) has only the trivial solution, so $T(\mathcal{B})$ is linearly independent.

(b). Suppose that T is invertible and $\mathcal{B}$ is a basis of $\mathbb{R}^n$. How are k, n, m related? Show that $T(\mathcal{B})$ is a basis of $\mathbb{R}^m$.

Solution: Since $\dim \mathbb{R}^n = n$ and any basis of a subspace has the same size, $k = n$. In addition, since T is linear and invertible, $n = m$ (see Corollary 2.7), so $k = n = m$. By part (a), we know that $T(\mathcal{B})$ is linearly independent set of m vectors in $\mathbb{R}^m$. By the basis theorem, it is a basis of $\mathbb{R}^m$.

Problem 8 Solution

rotation by 45° counterclockwise: $\mathbf{x} \mapsto A\mathbf{x}$, where

$$A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}, \quad \theta = 45^\circ.$$

reflection across the y-axis: $\mathbf{x} \mapsto B\mathbf{x}$, where

$$B \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

$T(\mathbf{x}) = B(A\mathbf{x})$ so has the standard matrix

$$BA = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}.$$

Problem 9 Solution

(a)

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 5 & 1 & 3 & 2 \end{pmatrix}$$

Inversions:

$$[1,3], [1,4], [1,5], [2,3], [2,4], [2,5], [4,5].$$

so $\text{inv}(\sigma) = 7$. $\Rightarrow \text{sgn}(\sigma) = (-1)^7 = -1$.

(b) To get σ, start from

$$(1\ 4) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 2 & 3 & 1 & 5 \end{pmatrix}$$

$$\downarrow (2\ 5)$$

$$\begin{pmatrix}1&2&3&4&5\\4&5&3&1&2\end{pmatrix} = (2\ 5)\circ(1\ 4)$$

$$\downarrow (1\ 3)$$

$$\sigma = \begin{pmatrix}1&2&3&4&5\\4&5&1&3&2\end{pmatrix} = (1\ 3)\circ(2\ 5)\circ(1\ 4)$$

So $\sigma = (1\ 3)\circ(2\ 5)\circ(1\ 4)$ is a composition of 3 transpositions. Thus, $\text{sgn}(\sigma) = (-1)^3 = -1$.

RMK : the representation is not unique.

Problem 10 Solution

(a)

$$\sigma = \begin{pmatrix}1&2&3&4&5\\4&3&1&5&2\end{pmatrix}$$

Inversions:

$$[1,2],[1,3],[1,5],[2,3],[2,5],[4,5].$$

so $\text{inv}(\sigma) = 6$. $\Rightarrow \text{sgn}(\sigma) = (-1)^6 = 1$.

(b) To get σ, start from

$$(1,4) = \begin{pmatrix}1&2&3&4&5\\4&2&3&1&5\end{pmatrix}$$

$$\downarrow (1,5)$$

$$\begin{pmatrix}1&2&3&4&5\\4&2&3&5&1\end{pmatrix} = (1,5)\circ(1,4)$$

$$\downarrow (1,2)$$

$$\begin{pmatrix}1&2&3&4&5\\4&1&3&5&2\end{pmatrix} = (1,2)\circ(1,5)\circ(1,4)$$

$$\downarrow (1,3)$$

$$\sigma = \begin{pmatrix}1&2&3&4&5\\4&3&1&5&2\end{pmatrix} = (1,3)\circ(1,2)\circ(1,5)\circ(1,4)$$

So $\sigma = (1,3)\circ(1,2)\circ(1,5)\circ(1,4)$ is a composition of 4 transpositions. Thus, $\text{sgn}(\sigma) = (-1)^4 = 1$.

(c) Let $\tau = (1,4) = \begin{pmatrix}1&2&3&4&5\\4&2&3&1&5\end{pmatrix}$

Consider $(\tau\circ\sigma)(x)$ for each x:
$(\tau\circ\sigma)(1) = \tau(\sigma(1)) = \tau(4) = 1$
$(\tau\circ\sigma)(2) = \tau(\sigma(2)) = \tau(3) = 3$
$(\tau\circ\sigma)(3) = \tau(\sigma(3)) = \tau(1) = 4$
$(\tau\circ\sigma)(4) = \tau(\sigma(4)) = \tau(5) = 5$
$(\tau\circ\sigma)(5) = \tau(\sigma(5)) = \tau(2) = 2$

Thus,

$$(1,4)\circ\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 3 & 4 & 5 & 2 \end{pmatrix}$$

Consider $(\sigma\circ\tau)(x) = \sigma(\tau(x))$ for each x:
$(\sigma\circ\tau)(1) = \sigma(\tau(1)) = \sigma(4) = 5$
$(\sigma\circ\tau)(2) = \sigma(\tau(2)) = \sigma(2) = 3$
$(\sigma\circ\tau)(3) = \sigma(\tau(3)) = \sigma(3) = 1$
$(\sigma\circ\tau)(4) = \sigma(\tau(4)) = \sigma(1) = 4$
$(\sigma\circ\tau)(5) = \sigma(\tau(5)) = \sigma(5) = 2$
Thus,

$$\sigma\circ(1,4) = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 3 & 1 & 4 & 2 \end{pmatrix}$$

So $(1,4)\circ\sigma \neq \sigma\circ(1,4)$

(d) $\operatorname{sgn}((1,4)\circ(2,3)\circ(2,5)) = (-1)^3 = -1$
$\operatorname{sgn}(\sigma\circ(1,4)\circ(2,3)\circ(2,5)) = \operatorname{sgn}(\sigma)\cdot\operatorname{sgn}((1,4)\circ(2,3)\circ(2,5)) = 1\cdot(-1) = -1$

(e) $(1,5) = (1,2)(2,3)(3,4)(4,5)(3,4)(2,3)(1,2)$

Problem 11 Solution

Using the elimination method, we obtain:

$$A = \begin{bmatrix} 1 & 2 & a \\ -2 & 4a & 2 \\ a & -2 & 1 \end{bmatrix} \xrightarrow{R_2+2R_1, R_3-aR_1} \begin{bmatrix} 1 & 2 & a \\ 0 & 4a+4 & 2+2a \\ 0 & -2-2a & 1-a^2 \end{bmatrix} = B$$

Let us consider two cases.

Case 1: $a = -1$. Then the matrix B is equal to

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Therefore, B (and hence A) has rank 1.

Case 2: $a \neq -1$. Then we divide the second and the third rows of B by $4a+4$ and $-2-2a$ respectively:

$$\begin{bmatrix} 1 & 2 & a \\ 0 & 4a+4 & 2+2a \\ 0 & -2-2a & 1-a^2 \end{bmatrix} \xrightarrow{R_2/(4a+4), R_3/(-2-a)} \begin{bmatrix} 1 & 2 & a \\ 0 & 1 & \frac{1}{2} \\ 0 & 1 & \frac{(1-a)(1+a)}{-2-2a} \end{bmatrix} \xrightarrow{R_3-R_2} \begin{bmatrix} 1 & 2 & a \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & \frac{a-2}{2} \end{bmatrix} = C.$$

Let us again consider two cases.

Case 2a: $a = 2$. Then

$$\begin{bmatrix} 1 & 2 & 2 \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & \frac{a-2}{2} \end{bmatrix} = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix}$$

has rank 2.

Case 2b: $a \neq 2$. Then

$$\begin{bmatrix} 1 & 2 & a \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & \frac{a-2}{2} \end{bmatrix} \xrightarrow{R_3/(a-2)/2} \begin{bmatrix} 1 & 2 & a \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 1 \end{bmatrix}.$$

The last matrix has rank 3.